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AI Based Energy Theft Detection in Smart Meters

Bachelor Thesis

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This is to certify that:

(i) The thesis comprises only my original work toward the Bachelor Degree

(ii) Due acknowledgement has been made in the text to all other material used

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5 January, 2026

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Abstract

Smart grids are energy networks that automatically monitor energy flows and adjust to changes in supply and demand. When paired with smart meters, they provide real-time information on energy usage to utility providers. While smart grids have become widely adopted in recent years and improve grid efficiency, they also raise serious privacy concerns. Sending raw energy consumption data from households or companies to utility providers creates risks: this data can be attacked or intercepted, revealing sensitive information about consumers' activities and enabling threats like electricity theft. These risks exist because smart grids operate in a centralized environment where all data flows through a single point. To preserve consumer privacy while maintaining the benefits of smart grids, this study introduces federated deep anomaly detection (FDAD), a decentralized framework that detects anomalies collaboratively and securely without exposing raw consumption data. In this approach, a deep anomaly detection model, multi-layer perceptron (MLP), is trained on each consumer's data locally. The learned parameters are sent to a central utility provider for aggregation instead of raw data. federated learning (FL) occurs over several communication rounds, with anomaly prediction performed during the last round. The proposed framework is tested using real smart meter data from various households. The performance of FDAD for each household is compared to a standalone MLP model trained on the same household's data with the same number of epochs used during FL, as well as the optimal number of epochs. Experimental results show that FDAD greatly outperforms standalone MLP models trained for a limited number of epochs; however, when compared to MLP trained on the optimal number of epochs, FDAD shows slightly lower performance. Although the proposed approach does not exceed the optimal standalone training, it provides significant benefits in privacy protection, scalability, and decentralized deployment with promising performance. These findings indicate that FL offers a practical and effective solution for detecting energy theft in smart meters, giving a balance between detection performance, data privacy, and regulatory requirements.

Keywords: *FL*, *MLP*, *Smart meters*, *Standalone training*

Contents

Abstract **4**

1 Introduction **7**

Introduction **7**
1.1 Background 7
1.2 Motivation 7
1.3 Research Questions 8
1.4 Research Objectives 9
1.5 Thesis Organization 9

2 Related Works **10**
2.1 Smart Meters Applications 10
2.2 Traditional ML for Theft Detection 11
2.3 Deep Learning For Anomaly Detection 12
2.4 Unsupervised Anomaly Detection Methods 13
2.5 Smart Meters Security Risks 14
2.6 Federated Learning 15

3 Preliminaries and System Modeling **19**
3.1 Preliminaries 19
3.1.1 Machine Learning: 19
3.1.2 Isolation Forest 20
3.1.3 Multi-Layer Perceptron 20
3.2 System Model 21
3.3 Performance Evaluation Metrics 22

4 Methodology **24**
4.1 Offline Mode 24
4.2 Online Mode 26

5 Performance Evaluation **28**
5.1 Dataset Description 28
5.2 Experimental Setup and Results 28
5.2.1 *Consumption Analysis:* 28

5.2.2 *Data Labeling* 30
5.2.3 *Local MLP Training:* 32
5.2.4 *Federated Learning* 33
5.3 Comparison and Analysis 35

6 Conclusion and Future Works **37**

References **3**

Chapter 1

Introduction

This chapter provides background information about energy theft in smart meters, explains the motivation behind this study along with the proposed method, research questions, and objectives of this study.

1.1 Background

Over the past few years, energy prices have risen significantly, motivating researchers to design more sustainable homes and energy-efficient systems to reduce costs. However, this increase in energy prices has also driven some individuals, companies, and organizations to engage in energy theft as a means of lowering their expenses. Energy theft can be in the form of fraud (meter tampering), stealing (illegal connections), billing irregularities, and unpaid bills. Energy theft can lead to huge financial losses to individuals, companies, and the government. To address these challenges, modern power systems are evolving into smart meters. Smart meters provide fine-grained measurements of household energy usage in real time. They continuously measure and record energy consumption and have a two-way communication system with the utility provider, which is the organization responsible for supplying electricity to consumers. They send energy consumption information to the utility provider, as well as receive instructions and updates from it automatically. As illustrated in Figure 1.1, this approach can save costs as the readings are sent accurately to the utility provider and in real time, avoiding any computational or log errors that can occur if data are gathered manually. It can also help detect energy theft through monitoring any unusual behavior or anomalies and alert home owners or company owners immediately.

1.2 Motivation

Smart meters are advanced in their ability to monitor energy consumption and their ability to have two-way communication between homes and utility providers in real time. Smart meters can also detect energy theft using machine learning models that can accurately detect whether a behavior is considered suspicious or not. However, applying machine learning to smart meters requires centralized data processing, which introduces significant security risks. To overcome

this challenge, this study proposes a federated deep anomaly detection (FDAD) approach that uses the FedAvg algorithm combined with a multi-Layer perceptron (MLP) model to accurately detect energy theft through anomaly detection. The proposed framework protects user privacy by training the MLP model directly on smart meter devices installed locally without ever sharing raw consumption data with the utility provider, while still achieving high detection accuracy.

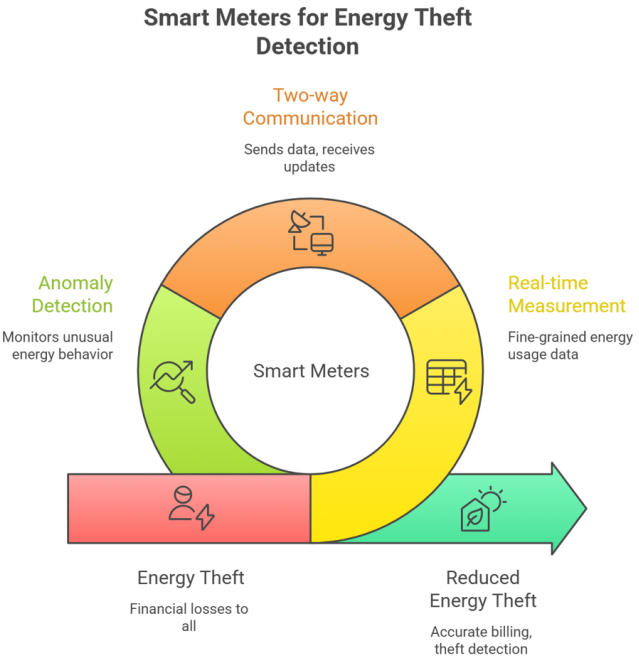


Figure 1.1: Smart Meters can directly reduce energy theft

1.3 Research Questions

1. How can Federated Learning be applied to smart meters for anomaly detection without exposing raw data?

2. Can a FDAD model achieve comparable performance to a standalone MLP trained on the same household data?

3. What is the trade-off between privacy preservation and anomaly detection accuracy in FDAD?

4. What are the benefits and limitations of using federated deep learning for real-time energy anomaly detection in smart grids?

1.4 Research Objectives

The main objectives of this thesis are:

1. To identify and apply a deep learning model that accurately detects anomalies.
2. To apply Federated Learning to enable privacy-preserving and scalable anomaly detection across distributed smart meters.
3. To evaluate the performance of the proposed federated approach in terms of detection accuracy, data privacy, and computational efficiency.

1.5 Thesis Organization

The remainder of this thesis is organized as follows. Chapter 2 reviews the relevant literature. Chapter 3 presents the preliminaries and foundational concepts necessary for understanding the proposed approach. Chapter 4 describes the methodology and outlines the steps taken to achieve the research objectives. Chapter 5 presents the results of the local model training and federated learning approaches, along with a comparative analysis and the evaluation metrics used. Finally, Chapter 6 concludes the thesis and discusses potential directions for future work.

Chapter 2

Related Works

To place this work on context, several related studies on energy theft detection and anomaly detection are examined in this section as well as tables 2.1 and 2.2 provide a simple summary of all papers discussed below.

2.1 Smart Meters Applications

Smart meters are devices that capture real-time energy consumption data. This allows for better monitoring and management of electricity usage. They have developed rapidly in recent years and are now used in various areas, from consumer-level monitoring to grid management. Dominicis et. al [1] highlighted how smart meters can save energy and let consumers achieve a more sustainable lifestyle. They applied a simple experiment where they investigated the long-term effect of of a normative feedback system in which residents' electricity consumptions are compared to that of similar or neighboring households. Each household that participated in that experiment had three lights: green (the household is consuming less energy than others), yellow (the household is consuming the same amount of energy), and red (the household is consuming more energy). This methodology is built on data gathered in real-time using smart meters. The results of this experiment show an overall energy reduction in this neighborhood of approximately 4.57% less energy consumption over two years. Moreover, Raza et al. [2] provide a detailed overview of smart meters in business intelligence applications. Their work shows how smart meters gather data that can be analyzed across IoT and computational layers. This includes local devices as well as fog and cloud computing, which helps extract useful insights. These insights support predictive modeling, anomaly detection, and improvement of energy distribution. This demonstrates the many ways smart meter data can be used beyond basic consumption tracking. Wang et al. [3] review smart meter data analytics. They highlighted that smart meters are used in applications such as load analysis, load forecasting, load management, abnormal consumption detection, consumer segmentation, and demand response. A summary of the advantages of smart meters usage is illustrated in Figure 2.1 below.

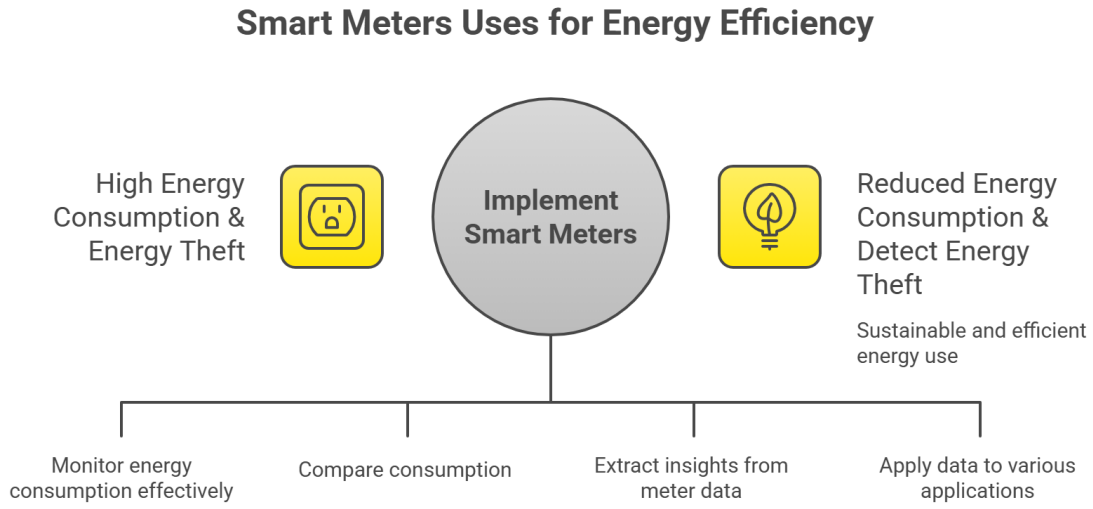


Figure 2.1: Smart meters advantages

2.2 Traditional ML for Theft Detection

Detecting theft is crucial for reducing financial losses. Therefore, early researchers have examined various statistical and machine learning methods to identify unusual or fraudulent behaviors. In smart grids, non-technical losses (NTL) come from electricity theft, billing mistakes, and fraud, resulting in significant challenges for utility providers. Early studies mainly used traditional machine learning techniques along with careful feature engineering. For instance, Abro et al. [4] looked into detecting electricity theft by testing seven classical machine learning classifiers with different data balancing methods and training-test split ratios. Their extensive experiments revealed that Random Forest paired with Random Over Sampling and a 70-30% split produced the best overall results, achieving a precision of 0.990 with significant improvements. Likewise, Hong et al. [5] used a Support Vector Machine (SVM) model to spot electricity theft in electric vehicle charging stations, achieving excellent accuracy and recall. These studies highlight the effectiveness of traditional machine learning models when used with suitable preprocessing and class balancing strategies. However, these methods usually rely on manually engineered features and struggle with the complex temporal or spatial relationships in consumption data. Tehrani et al. [6] suggests a machine learning solution based on Decision Tree, Random Forest, and Gradient Boosting classifiers. The classifiers were tested on datasets collected from smart meters installed in 114 single-family homes. To reduce the impact of class imbalance in daily electricity consumption, the authors used various preprocessing methods, which included feature engineering and k-means clustering. The outcomes demonstrated that classification on the clusters was far more effective, with Gradient Boosting being the best among all methods.

Moreover, Recent studies confirmed that classical machine learning techniques are still very

effective for real-world electricity theft identification, particularly when there are no smart meter installations. In one research, the proposed technique covered post-paid customers with conventional meter installations. In this setup, the identification of electricity theft heavily depends on manual survey processes. The researchers proposed a machine learning technique to optimize the target operation (TO) process to determine the list of potential customers for identification through the analysis of their consumption patterns for each months. A number of classifiers were considered, which included decision tree, naive bayes, k-nearest neighbor, logistic regression, random forest, and a deep neural network. Of the classifiers suggested, random forest performed the best, even for unbalanced data. However, to further upgrade the accuracy of the proposed technique, the researchers presented an alternative sequential identification strategy to further apply multiple levels of filtering based on the categorized theft frequency, hence minimizing the number of false positives. In this work, the proposed Random Forest and KNN classifier produced an accuracy of 0.89 and an AUC value of 0.89 [7].

2.3 Deep Learning For Anomaly Detection

More recently, deep learning based anomaly detection has emerged as a strong alternative, especially for complex data. Nirmal et al. [8] suggested a hybrid CNN-AdaBoost model, using convolution neural networks for feature extraction and AdaBoost for classification. This hybrid model shows a trend of blending deep feature learning with classical ensemble classifiers. Similarly, Hasan et al. [9] proposed a CNN-LSTM architecture, where CNN layers capture spatial patterns and LSTM layers model temporal relationships in electricity consumption data. While both approaches improved detection performance compared to traditional methods, they also increased computational complexity and relied on offline analysis, which limits their use for real-time applications. Elshenawy et. al [10] utilized a CNN-LSTM approach for anomaly detection; however, they have also focused on achieving state-of-the-art accuracy. To handle class imbalance, they have employed LoRAS data augmentation. Their approach achieved an accuracy of 97%, surpassing state-of-the-art results. Xia et. al [11] have also used CNN for anomaly detection and handled class imbalance using focal loss. Experimental results demonstrate that compared to other state-of-the-art deep learning classifiers, this method has better performance, with 97.08% of mean average precision (MAP@100) and 83.61% of AUC, which verifies the effectiveness of the proposed method. Massarani et. al [12] proposed a two step pipeline where they combined data condensation techniques, such as principal component analysis (PCA) and K-means clustering to extract representative consumption patterns, with an ensemble learning approach: one-class support vector machine (OCSVM) and the gaussian mixture model (GMM). Results show that the proposed sensor based detection framework achieves superior performance, with an accuracy of 88.45% and a false alarm rate of just 13.85%, while reducing training time by over 75%. While CNN and ensemble methods have demonstrated strong performance, sequence oriented models can better capture temporal dependencies in energy consumption data. Javaid et. al [13] proposed an efficient state-of-the-art approach: autoencoder-bidirectional GRU model. Autoencoder is used for feature extraction which captures essential electricity consumption patters, and BiGRU network leverages long-term memory to classify anomalies with minimal false positives.This study has

also focused on class balancing by employing synthetic theft attacks. The model is evaluated using real datasets leveraged from state grid corporation of China. The results demonstrate that the proposed approach effectively balances high accuracy (90%) with a low false positives (10.2%).

2.4 Unsupervised Anomaly Detection Methods

Beyond supervised theft detection, anomaly detection has been studied as a way to find unusual patterns in unlabeled or weakly labeled datasets. Early unsupervised anomaly detection methods largely relied on geometric and density-based principles. Eskin et al. [14] put forward a geometric framework where anomalies are detected based on their distances and density relationships in feature space, which laid the groundwork for many later methods. Building on similar ideas, Breunig et al. [15] introduced the local outlier factor (LOF) in anomaly detection, which assigns a degree of abnormality to each instance instead of a simple binary label. This local density perspective allows LOF to detect anomalies that may be hidden in areas with varying data density, influencing many locality-aware anomaly detection techniques. Tree-based anomaly detection methods later became popular due to their scalability and few assumptions about data distribution. Isolation Forest offered a different strategy by isolating anomalies through random partitioning instead of modeling normal behavior. While effective and computationally efficient, standard Isolation Forest has biases from axis-aligned splits. To solve this, the extended isolation forest (EIF) was introduced, allowing splits with random orientations, enhancing the robustness and consistency of anomaly scores without raising computational costs. Further variants, like SA-iForest, add adaptive mechanisms to better manage subtle anomalies and class imbalances, making them particularly relevant for detecting electricity theft [16]. Kernel-based methods, like one-class SVM, offer a different perspective. They learn a boundary that encloses normal data in feature space. These methods are similar to geometric and density-based approaches, but they rely on kernel functions and optimization constraints. While they work well for moderately sized datasets, their computational complexity can limit their use in large smart grid situations [5].

Clustering based approaches have also been explored for anomaly detection. Qi et. al [17] introduced a novel unsupervised data-driven method for electricity theft detection which uses real smart meter data, discrete wavelet transform feature extraction, and fuzzy-c means clustering. Anomaly scores are computed from cluster membership degrees to distinguish normal and anomalous behavior. The proposed method is compared with a set of baselines including state-of-the-art methods using smart meter data of both business users and residential users. The comparison results indicate that the proposed method achieves significantly better detection performance than all baseline methods. However, the method utilized in this study can only maintain the detection performance when being applied in a shorter time frame, it cannot detect daily tampering behaviors of a user. In contrast to their earlier work relying on clustering and observer meter analysis, Qi et. al [18] proposed an unsupervised anomaly detection using an ensemble method. The ensemble approach combines two unsupervised anomaly detection algorithms: k-nearest neighbors (KNNs) and gaussian mixture models (GMM), with a correlation analysis technique based on mutual information. The anomaly scores produced

by each method are combined using a weighted strategy determined by K-means clustering. Experimental results on residential and commercial datasets show that the ensemble method outperforms individual detectors and existing unsupervised approaches.

2.5 Smart Meters Security Risks

The detailed nature of smart meter data allows for a thorough analysis of energy use. However, it also raises significant privacy issues since individual usage patterns can expose sensitive information about household behavior. Wang et. al [19] have looked into security and privacy risks in smart grids, especially focusing on smart meters as vulnerable points. Survey studies point out vulnerabilities that stem from central data collection and stress the need for designs that consider privacy. Tanimoto et. al [20] have highlighted how smart grids gateways raise security risks. The authors have extracted 37 risk factors using risk breakdown structure (RBS) method, clarified the tendency and issues relating to each risk, and proposed new countermeasures such as information security policies, ISMS certification, PKI deployment, and privacy protection mechanisms. Other research suggests cryptographic solutions like privacy-preserving data aggregation and differential privacy to limit information leaks while still allowing for grid monitoring [21], [22]. Cao et. al [23] presents a successive framework for secure and accurate smart grid data storage. The framework combines parallel vision, used to verify and standardize meter readings, and cloud-blockchain fusion model, which stores accurate data with tolerable latency and throughput. This approach enhances reliability and security of smart grids. Existing security methods can significantly improve the protection of smart meter data as shown in Figure 2.2; however, they usually focus on the protection of data storage or transmission. On the other hand, federated learning preserves data privacy during machine learning stage. The next section explores methods researchers have proposed for data preservation during training using federated learning approach.

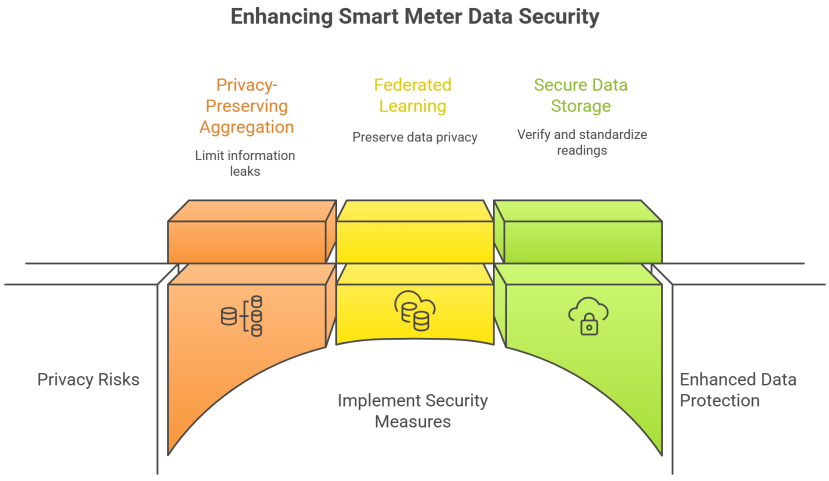


Figure 2.2: Secure solutions for data preservation

2.6 Federated Learning

Federated learning has become a strong method for collaborative machine learning. It allows for this collaboration without needing centralized access to raw data. This approach addresses privacy concerns and isolated data issues across various fields. Sheller et al. [24] show how FL applies to medicine. Their results show that FL across 10 institutions results in models reaching 99% and surpasses centralized approaches, while preserving data privacy. While FL enables privacy-preserving model training, its practical use is challenged by differences in data distributions across institutions. FL has also been successfully applied in smart grids. As discussed earlier, smart grids systems send large volumes of data to a centralized cloud storage that is in risk of attacks, can reveal sensitive energy consumption data, and increased communication overhead. To overcome these challenges, Su et. al [25] proposed an edge-cloud integrated FL framework for privacy-preserving and communication-efficient energy data analysis in smart grids. They have also highlighted a challenge that usually occurs in energy consumption data during FL which is non-IID. Non-IID data refers to data distributions that vary across users, where local datasets differ in size, characteristics, and statistical properties. To solve this challenge, the authors introduced a local data evaluation system that checks the quality and usefulness of each user's data before it adds to the global model. They also developed two optimization problems. One problem targets energy data owners (EDOs) and focuses on fair incentives and participation efficiency. The other problem is for energy service providers (ESPs) and emphasizes model accuracy and system cost. To promote participation and tackle uncertainty in users' private information, they included a two-layer deep reinforcement learning-based incentive mechanism. This system adjusts rewards in real-time to improve user engagement and enhance overall system performance.

To tackle the challenges of heterogeneity in FL, Li et al. [26] suggest FedProx. This optimization framework builds on FedAvg to manage both statistical and system differences. FedProx adds a proximal term to stabilize local updates. It lets devices do varying amounts of work, which helps improve convergence stability in diverse networks. Their results show improvements in accuracy and convergence robustness compared to standard FedAvg. Gupta et al. propose FedGrid, a secure FL framework for energy optimization. It uses privacy-preserving predictive models to forecast renewable energy generation and electricity use. Their research shows that FL supports informed energy supply decisions while protecting user privacy, but practical deployment faces issues such as varied client participation, resource limitations in smart meters, and regulatory compliance [27]. Liu et. al [28] proposed a three-party privacy-preserving data aggregation (3PDA) approach where data privacy is preserved by masking each user's data using information from other users. Instead of relying on a single third party for data aggregation, the method allows users to collaboratively form a virtual aggregation group, such that individual electricity consumption values cannot be inferred while the final aggregated result remains accurate for utility analysis. Moreover, Zhao et al. [29] presented a smart and efficient Privacy-preserving Data Aggregation (PDA) solution for the fog-based smart grid. The solution utilizes edge/fog computing to enable the data to be locally processed with preserved privacy. The authors presented two schemes: PDA with Smart Pricing (PDA-SP); this allows the control center to calculate complicated aggregation statistics and provides diverse pricing options. PDA with Packing Method (PDA-PM); this optimizes the size of the en-

crypted data to improve computational and communications efficiency. The results revealed that the solution is secure against a number of threats, and the experiment results proved the solution to be lightweight and efficient. This solution presents an idea on how privacy preservation during aggregation can complement FL, facilitating the locally processing of private energy consumption information at the edge with reduced communications overhead and increased privacy. In addition, another approach considers reducing the computational overhead in resource-constrained smart meter devices while maintaining the confidentiality of data. The work presents a lightweight and privacy-enabling masking data spatial aggregation scheme involving only simple cryptographic primitives, including hash functions and exclusive XOR operations. This approach enables secure power consumption forecasting and billing for the smart grid environment, and the performance analysis show that the proposed method achieves lower computational overheads relative to existing data aggregation techniques [30].
Collectively, these studies indicate that FL is a flexible and privacy-conscious approach to enable CI in areas such as healthcare and smart grids, without needing further access to the original data. However, according to the studies and as illustrated in figure 2.3 below, FL, when used alone, is a perfect solution, instead needing be used in conjunction with other methods in order to provide a solution to other, more realistic, problems such as non-IID datasets, heterogeneity, incentive schemes, and secure aggregation. Solutions such as utilizing both edge and cloud resources, reinforcement learning-based incentive structures, secure aggregation protocols, and methods of private aggregations of sensitive healthcare data, show that multiple methods work together to develop efficient and privacy-ensured smart grid networks.

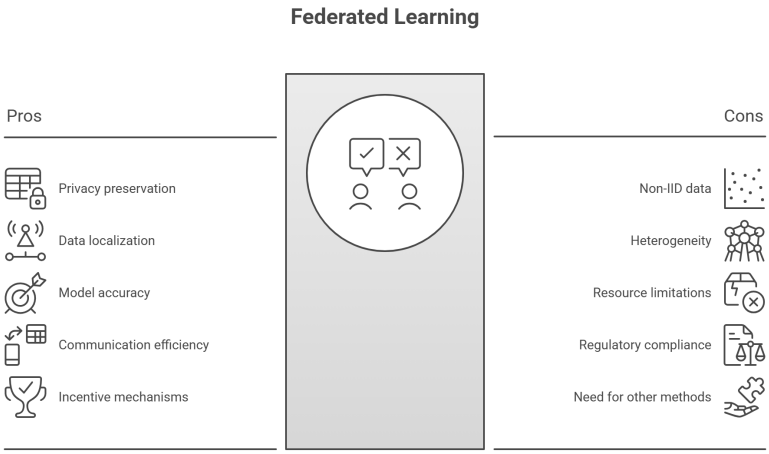


Figure 2.3: Federated learning pros and cons according to summarized papers

Table 2.1: Summary of Related Works in Energy Theft Detection and Smart Meter Applications (Part 1)

Authors	Methodology	Problem Addressed	Limitations
Smart Meter Applications			
Dominicis [1]	Normative feedback system	High energy consumption in residential buildings	Lacks generalization in population
Raza [2]	Advanced analytics, predictive modeling, generative AI	Enhance grid efficiency	Lacks specific theft algorithms and real-world validation
Wang [3]	Analyzes smart meter data	Understand smart meter analytics	No experiments conducted
Traditional Machine Learning			
Abro [4]	Random Forest with Random Over Sampling	NTL detection	Historical data only; not real-time
Hong [5]	SVM	Theft in EV charging stations	Struggles with complex temporal relationships
Tehrani [6]	Gradient Boosting + k-means clustering	Energy theft in smart meters	Cannot detect real-time tampering
Taruna [7]	Classical ML classifiers	Energy theft detection without smart meters	Depends on manual surveys; may not scale
Deep Learning			
Nirmal [8]	CNN-AdaBoost	Anomaly detection	High computation cost; limited dataset testing
Hasan [9]	CNN-LSTM	Spatial-temporal pattern recognition	High computational complexity; offline only
Elshennawy [10]	CNN-LSTM + LoRAS	Class imbalance	Focuses on accuracy over speed
Xia [11]	CNN	Anomaly detection with class imbalance	High computational cost
Massarani [12]	PCA + K-means + OCSVM/GMM	Improve data efficiency	Depends heavily on data quality
Javaid [13]	Autoencoder-Bidirectional GRU	Feature extraction	Needs synthetic theft attacks for training
Unsupervised Methods			
Eskin [14]	Geometric Framework	Theft detection for unlabeled data	High false alarm rates
Breunig [15]	LOF	Detect local outliers	Computational latency
Xu [16]	Extended Isolation Forest + SA-Forest	Isolation forest biases	Sensitive to hyperparameters
Qi [17]	DWT + Fuzzy-C Means	Unlabeled theft detection	Limited to short time frames
Qi [18]	Ensemble (KNN+GMM) with K-means weighting	Robustness across datasets	Complex weighted score management

Table 2.2: Summary of Related Works in Smart Grid Security and Federated Learning (Part 2)

Authors	Methodology	Problem Addressed	Limitations
Security Risks			
Wang [19]	Systematic survey	Identify smart meter vulnerabilities	Focuses on threats only
Tanimoto [20]	Risk analysis using RBS	Identify gateway security risks	Only proposes countermeasures
Li [28]	3PDA data aggregation	Protect meter data during aggregation	Covers aggregation only
Marks [22]	Differential privacy methods	Limit info leaks in grids	Mainly theoretical
Cao [23]	Cloud-blockchain framework	Secure data storage	Focused on storage; no real-time processing
Federated Learning & Security			
Sheller [24]	FL across 10 institutions	Collaborative ML without sharing data	Tested in healthcare; smart grid challenges remain
Su [25]	Edge-cloud FL for smart grid	Privacy-preserving, efficient data analysis	Non-IID data; heterogeneous devices limit performance
Li [26]	FedProx for heterogeneous FL	Convergence stability	Only addresses heterogeneity
Gupta [27]	FedGrid for energy optimization	Forecast energy use while preserving privacy	Practical deployment issues
Liu [28]	3PDA aggregation in FL	Protect individual meter data	Aggregation focus only
Zhao [29]	PDA-SP / PDA-PM using fog/edge	Local processing with privacy	Limited to fog/edge; may not scale
Gope [30]	Lightweight spatial data aggregation	Secure forecasting in low-resource devices	Low-computation focus; full FL not tested

Chapter 3

Preliminaries and System Modeling

3.1 Preliminaries

This section provides the foundational concepts and background knowledge necessary to understand the proposed FDAD. It covers an overview of machine learning and deep learning.

3.1.1 Machine Learning:

As illustrated in figure 3.1, machine learning is a subset or part of artificial intelligence (AI), where algorithms learn patterns on labeled and/or unlabeled data for the predictive and classification purposes of unseen data, known as the target. Labeled Data refers to the data in which

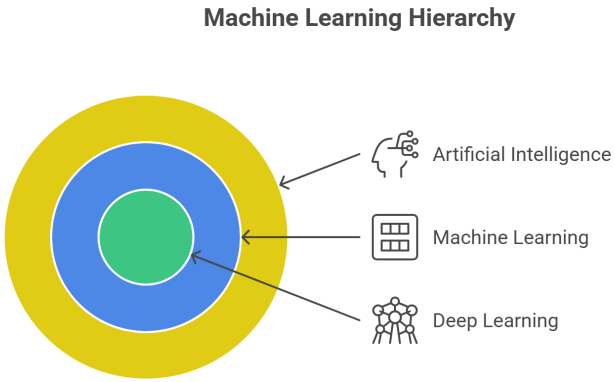


Figure 3.1: This figure shows the relationship between AI, ML, and deep learning.

the attributes are associated with the correct answers and outputs (targets), which is the best suited data for supervised learning algorithms, and is very helpful for the algorithm as it is responsible for the prediction of the output. Unlabeled data refers to data lacking a target value, which is usually used in unsupervised learning algorithms. Deep learning is a more advanced concept of machine learning that utilizes artificial neural networks that contains multiple layers, also referred to as a deep neural network. Deep learning can be used to extract complex features from datasets without requiring any human intervention of feature engineering. Deep

learning techniques are more successful in tasks like prediction, classification, and detecting anomalies.

3.1.2 Isolation Forest

The isolation forest is an unsupervised machine learning algorithm for anomaly detection. Basically, it isolates those data points that are substantially different from the normal observations by random partitioning. Since anomalies are a few and distinct, they are isolated faster than normal data, which allows the outliers to be found with low computational cost. It randomly selects a feature from the dataset and then splits the data at a random value in the range of the selected feature. By repeating this process, a collection of binary trees, called isolation trees, will be generated. Every internal node in the trees corresponds to a split, while every external node represents an isolated data point or a small number of points. The number of splits required to isolate a data point x in a tree is known as its *path length* $h(x)$. For a data point that ends in a leaf node containing n instances, the path length is described as:

$$h(x) = h' + c(n) \tag{3.1}$$

where h' is the number of splits , and $c(n)$ represents the average path length of an unsuccessful search in a binary search tree, used for normalization. Rather than using only one tree, the Isolation Forest uses an ensemble of T isolation trees, which are constructed independently using the technique of random feature selection and split values. Isolation trees are more robust to the anomaly detection task due to the diversity introduced by the randomness. The average path length of the data point can be calculated using the following formula:

$$\bar{h}(x) = \frac{1}{T}\sum_{i=1}^{T} h_i(x) \tag{3.2}$$

Taking the average path length as a reference, the model computes the anomaly scores based on the average path lengths; the shorter the average path lengths, the greater the chances of data points to be treated as anomalies. Finally, the results are classified using a threshold to decide which data points are normal or are to be tagged as anomalies based on the scores.

3.1.3 Multi-Layer Perceptron

Before explaining what a multi-layer perceptron is, there is a need to define what a neural network is. A neural network is a type of machine learning model that mimics the human brain, has the ability to learn from complex patterns, and make decisions on its own. It comprises two parts, neurons, which are nodes, which take inputs, and connections. Nodes are connected to one another and convey information, regulated by weights and biases. It further has three types of layers: input layer, which takes inputs where the input nodes receives input data; this represents features of a dataset, hidden layer, where most calculations are performed; they are considered to be the core of a neural network, they can consist of many layers, and an output layer, which are outputs or results obtained by performing operations on the input data. The learning process involves two steps. Step one is the forward propagation step, in which

the data moves from the input layer to the output layer via all the hidden layers. During this process, each input is multiplied by a corresponding weight. The weight defines the power or significance of the input. Secondly, the results obtained from the multiplication are added together to get the overall information. Finally, a bias is added to the overall information to give the overall output. The role or significance of this step involves the calculation of the weighted combination of the input variables in order to learn the relationship between the input and output variables. It can be mathematically represented as:

$$z = w_1x_1 + w_2x_2 + \ldots + w_nx_n + b \tag{3.3}$$

The output of the linear transform are then passed to an activation function. The activation function helps the system learn complex patterns by incorporating a non-linear element into the process. ReLU, sigmoid, and tanh activation functions are most suitable for deep learning networks.After the forward pass, the performance of the neural network on the current configuration of the weights is evaluated to determine how good it is, using the loss function, which compares the actual output to the output/result obtained. Next, backpropagation occurs which involves the network calculating the loss or the amount of error within the results produced. Then, the network calculates the gradient of loss with regard to each weight and bias within the network, indicating the strength or intensity of the impact or contribution towards loss, which can be described as:

$$w_{\text{new}} = w_{\text{old}} - \eta \frac{\partial L}{\partial w} \tag{3.4}$$

where η is the learning rate and $\frac{\partial L}{\partial w}$ is the gradient of the loss with respect to the weight. After computation of the gradients, the weights and biases are updated using an optimization algorithm, possibly stochastic gradient descent. The weights are changed in the direction opposite to that of the gradient to reduce loss. The amount by which each weight changes during an update depends on the learning rate. After carrying out a cycle consisting of forward propagation, computation of loss, backpropagation, and weight update, this cycle is repeated a number of times on the dataset. With time, the loss becomes lower, and predictions improve.
MLP is an artificial neural network model that can have several layers of neurons, typically consisting of three or more layers, and it forms the basic unit for the development of deep learning algorithms. MLP has fully connected/dense layers, meaning that every neuron within one layer connect to every neuron of the layer that follows it. These layers are essential for the model to learn non-linear relationships within the features of the data. This architecture ensures that the network can capture patterns and dependencies within the data, making it particularly effective for tasks that require understanding multidimensional relationships.

3.2 System Model

Figure 3.2 illustrates the system model that is utilized in this study. It follows a hierarchal cyber-physical architecture, and contains three main layers:

1. *Main Cyber-Physical Infrastructure (Main CPI)*: This level symbolizes the global infrastructure, which comprises a set of energy communities. Each energy community is administered by its respective Regional cyber-physical infrastructure.

2. *Regional Cyber-Physical Infrastructure (Regional CPI)*: Each regional CPI is responsible for monitoring the group of consumers/clients belonging to an energy community. The Regional CPI has the task of collecting, processing, as well as analyzing the consumers' consumption-related energy information. The proposed work will focus on this level that will act as the central server accepting the learned information from consumers as opposed to raw information regarding energy theft.

3. *Consumers (Homes)* : Each home in a regional CPI is equipped with multiple smart devices or IoT sensors that measure energy consumption. These devices send the collected data to the regional CPI through a gateway every N minutes in real-time. During this study, each consumer trains its data locally for anomaly using a deep learning model, and communicates with the regional CPI for n communication rounds for a real-time energy theft detection.

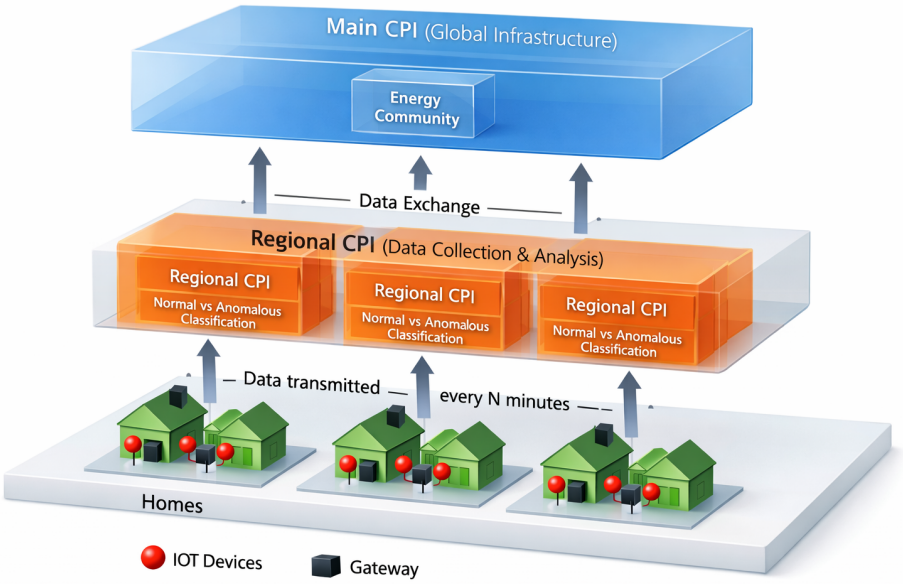


Figure 3.2: System model

3.3 Performance Evaluation Metrics

In order to effectively test the performance of the new approach, a set of performance measures is employed. In anomaly detection datasets, anomalies generally tend to form a small percentage of the overall data. Therefore, relying on the performance of the approach using the accuracy measure alone becomes unreliable. The accuracy formula measures the percentage of successfully made predictions out of the total prediction as:

$$Accuracy = \frac{True_Positives + True_Negatives}{Total} \tag{3.5}$$

In this regard, if a dataset had 98% normal points and 2% points that are anomalies, it means that the model was capable of reaching a maximum accuracy of 98% if it had marked all points as normal. Therefore, other metrics alongside accuracy are utilized, such as precision, recall, and F1-Score. First, precision measures the proportion of true anomalies out of all predicted anomalies and shows how many detected anomalies are actually correct. It is calculated as:

$$Perceison = \frac{True_Positives}{True_Positives + False_Positives} \tag{3.6}$$

Second, the recall metric indicates how many true anomalies the model successfully detected. It is identified as:

$$Recall = \frac{True_Positives}{True_Positives + False_Negatives} \tag{3.7}$$

Next, F1-Score represents the harmonic mean of precision and recall. It is given by:

$$F1_Score = 2 \times \frac{Precision \times Recall}{Precison + Recall} \tag{3.8}$$

Binary cross entropy (BCE) loss function is also employed during the local training of the MLP on each epoch. BCE calculates the difference between the actual class labels that are either 0 or 1 and the predicted class probabilities given by the model. The smaller the value of the BCE loss function, the better the predicted class labels match the actual labels. Mathematically, it is given by:

$$\text{BCE} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log\left(p_i\right) + \left(1 - y_i\right)\log\left(1 - p_i\right)\right] \tag{3.9}$$

Where N is the number of observations, y_i is the actual binary label (0 or 1) of the ith observation, and p_i is the predicted probability of the ith observation being 0 or 1.

Chapter 4

Methodology

In this chapter, the steps of the proposed FDAD approach is described. The methodology proposed has two major modes: the offline mode and the online mode.

4.1 Offline Mode

Offline mode is the phase where all data preparations before anomaly detection using FL occurs. Relating to the CPI discussed earlier in chapter 3, offline phase occurs in the consumer level, before sending any parameters to the regional CPI. As illustrated below in figure 4.1, It comprises of five main steps:

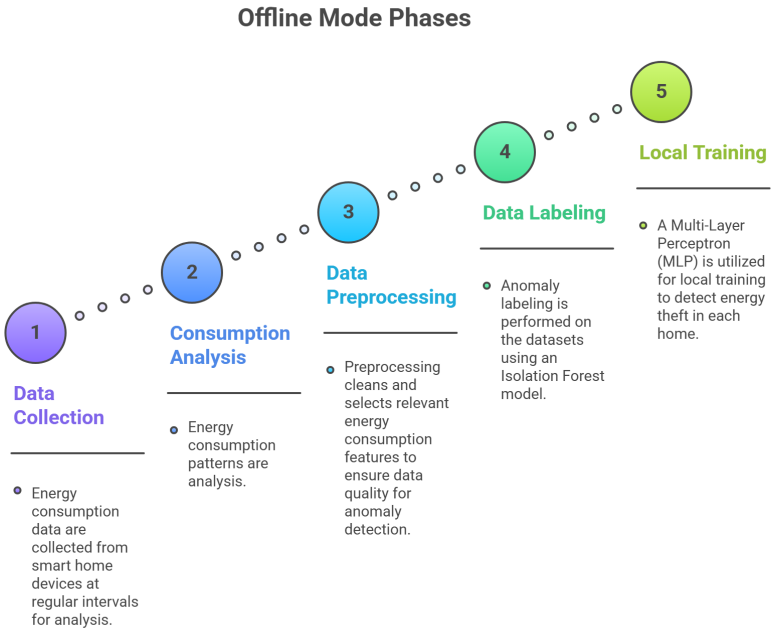


Figure 4.1: Offline mode steps

1. *Data Collection:* Energy consumption data are collected from smart meters installed in different households in real-time. The devices are designed to record consumption values and send these collected data to be analyzed.

2. *Consumption Analysis*: In this step, the consumption of energy is analyzed in order to understand the way the data behaves before the process of preprocessing. For example, it involves the analysis of the energy patterns of consumption on a weekly, monthly, or yearly basis, or finding the peak hours of consumption or the way a device consumes energy. It provides a basis of data on the way the energy is consumed based on the understanding gained.

3. *Data Preprocessing*: Data preprocessing involves converting raw data to clean and consistent data. This affects anomaly detection algorithms. Data preprocessing involves checking the missing and noisy data and handling them. It also involves checking heterogeneous energy consumption data from different households, and keeping only similar features or devices so all consumers have the same features for preparation of Federated Learning.

4. *Data Labeling*: The approach adopted in this paper is a supervised deep learning technique. This technique is dependent on labeled data. The process of assigning labels to data is an important part of data labeling and is performed on data that is not labeled before. In this paper, an Isolation Forest approach is used for data labeling. The data is normalized before training the model on it. Normalization is used to assign an equal weight to all variables and eliminate the impact of anomalies. The Isolation Forest approach is used along with some critical parameters. The number of estimators and contamination provide an estimation regarding how many trees should be there in the forest and how many data points are anomalies. The data points are labeled as anomalies on the basis of these parameters.

5. *Local training:* Local training utilizes MLP model. The input layer takes a vector of features representing energy consumption devices as input. These inputs pass through fully connected hidden layers to learn energy consumption patterns. Each hidden layer applies a non-linear activation function, ReLU, which is represented as:

$$\text{ReLU}(z) = \max(0, z) \tag{4.1}$$

   In order to avoid overfitting and improve generalization, a dropout regularization layer is applied after the ReLU activation function. Dropout randomly drops a number of neurons on each training iteration, making the network learn robust and redundant feature representations. The output layer takes all the features learned in the hidden representation and maps it through a fully connected layer to a single value. Afterwards, a sigmoid activation function is applied:

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}} \tag{4.2}$$

This function outputs a probability value between 0 and 1, indicating the likelihood of a data point being anomalous. The overall forward propagation of the model can be expressed as:

$$\hat{y} = \sigma\left(W_3 f\left(W_2 f\left(W_1 x + b_1\right) + b_2\right) + b_3\right) \tag{4.3}$$

where x is the input feature vector, W_i and b_i represent the weight matrices and bias vectors of the network layers, respectively, and $f(\cdot)$ is the ReLU activation function with dropout applied during training. The complete model architecture used for local training is illustrated in Figure 4.2.

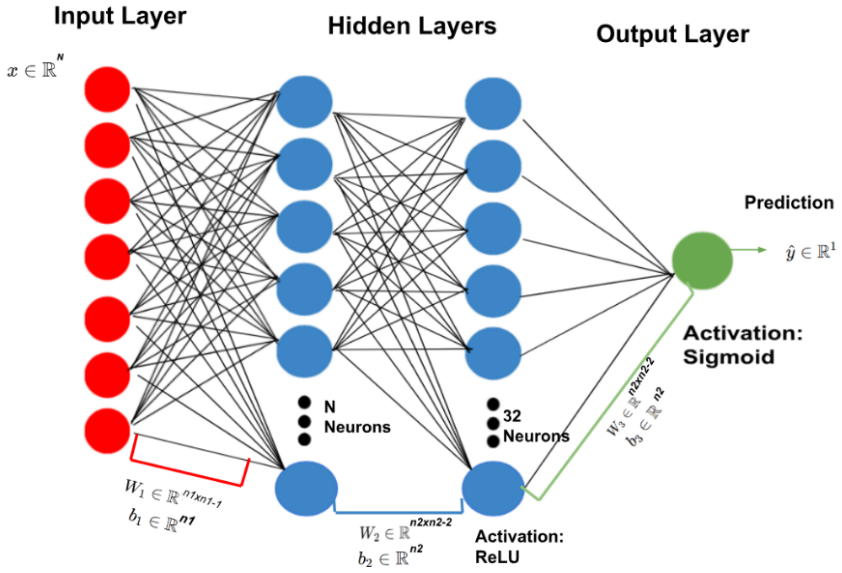


Figure 4.2: MLP Local model architecture

4.2 Online Mode

During the online mode, the federated learning framework is used to allow decentralized training across multiple homes (clients/consumers) while keeping data private. Each client randomly initializes the model parameters during the first round. Each client then trains locally on its training set using mini-batches of N samples. In the training process, a forward pass occurs. This is followed by calculating the loss using Binary Cross-Entropy (BCE). This method checks the difference between the predicted labels and the true labels. The model parameters are updated through backpropagation with the Adam optimizer. After training for N epochs, each client sends its updated model parameters to the central server. The server combines the received parameters from all clients using Federated Averaging (FedAvg). This is calculated as follows:

$$W_{\text{global}} = \frac{\sum_{i=1}^{N} n_i \cdot W_i}{\sum_{i=1}^{N} n_i} \tag{4.4}$$

where W_i represents the model weights from client i, n_i is the number of training samples at client i, and N is the total number of clients. This weighted average ensures that clients with larger datasets have proportionally greater influence on the global model. The aggregated global weights are then distributed back to all clients, and the process repeats for N rounds. The full system architectures is shown in Figure 4.3

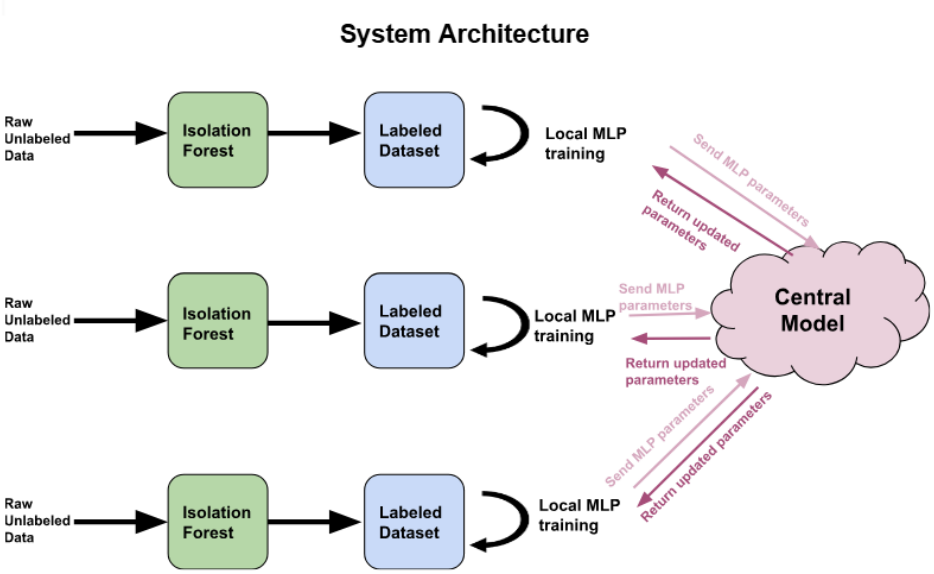


Figure 4.3: System architecture

Chapter 5

Performance Evaluation

5.1 Dataset Description

The dataset that is utilized in this study is obtained from Smart*. Smart* is a platform that continuously collects detailed electricity consumption data from seven residential buildings (Homes A, B, C, D, F, G, & H) [31]. These datasets are recorded over a span of three years (from 2014 - 2016) and can be used for multiple purposes beyond energy theft detection. In this work, data from Homes B, C, and F during the year 2014 are selected for analysis. Each home records electricity usage for various household devices at different time intervals. The datasets are unlabeled, which means that there is no a target feature, anomaly. After preprocessing, the resulting datasets provide energy consumption measurements at 30-minute intervals, covering the following devices: microwave, refrigerator, dishwasher, washer-dryer, lighting (aggregated across all lighting fixtures), and furnace. All measurements are expressed in kilowatts (kW). For each selected home, the processed dataset consists of 17,524 observations, enabling a detailed analysis of daily electricity consumption patterns for anomaly detection experiments.

5.2 Experimental Setup and Results

This section illustrates the FDAD approach detailed implementation using the smart dataset in 4 steps: consumption analysis, data labeling, local training, and federated learning.

5.2.1 *Consumption Analysis:*

As mentioned in Chapter 4, temporal energy consumption patterns are analyzed for an initial understanding of energy consumption in homes. The analysis of energy consumption patterns is presented through three perspectives: monthly average consumption, average daily consumption, and average hourly consumption patterns. As shown in Figure 5.1, each home has different peak energy consumptions measured in kilowatts. For instance, home B reaches peak energy consumption during the month of July, with an estimate of 4 KWs. Meanwhile, home C reaches its highest in February, with an average of 3.5 KWs. Home F has a very distinct behavior than homes B and C, reaching highest energy consumption during June, with

approximately 10 KWs. The patterns of the three homes reveals that there is a seasonal variation, where homes B and F consume the most energy during the summer; whereas, home C consumes the most during the winter time. This distinct patters between homes highlights the heterogeneous nature of energy consumption data, which occurs realistically. Figure 5.2 shows the average amount of daily energy consumption during a week. Home B and Home C have the highest consumption on Sundays, while Home F's peak consumption is on Mondays. These variations in consumption highlights the differences of consumption patterns in households. The increased amount of consumption at Home B & C on Sundays indicates higher occupancy of the house on the weekend with increased use of devices for entertainment, cooking, and house-hold chores. On the contrary, the peak at Home F on Monday could imply variations in work hours, probably due to home-based work. Finally, Figure 5.3 illustrates the average hourly energy consumption. While Home B and Home F have the peak point in energy consumption at 7 PM, Home C has its peak point at 9 PM. Peak consumption in energy occurred during the evening hours. This is because most individuals carry out most home chores when they get home from work. Therefore, they consume more energy during this particular period. This trend in the three houses indicates that any behavior that goes beyond these patterns might be considered irregular, possibly leading to energy theft.

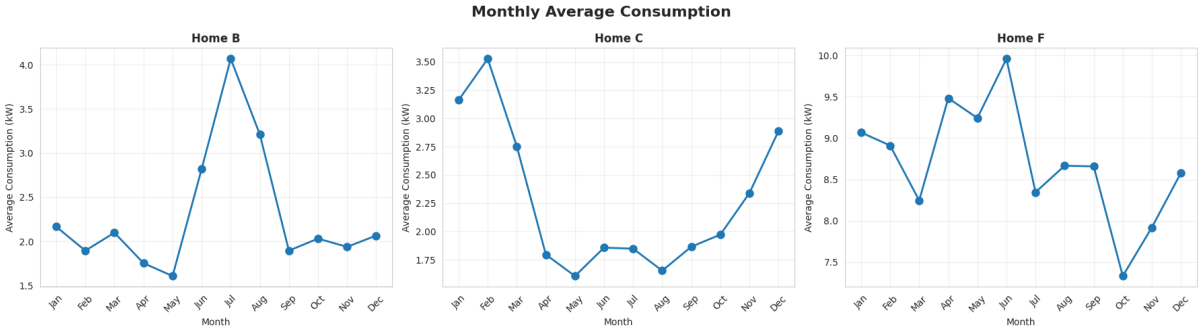


Figure 5.1: Monthly average energy consumption patterns for homes B, C, & F.

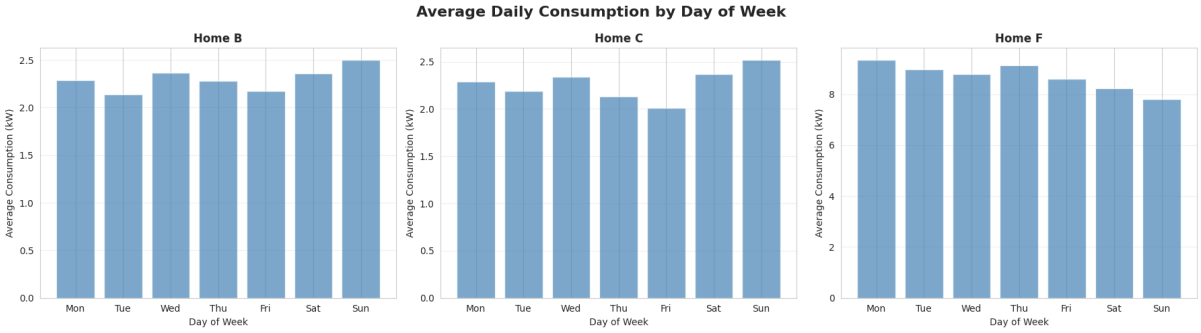


Figure 5.2: Average daily energy consumption patterns of all homes using histogram.

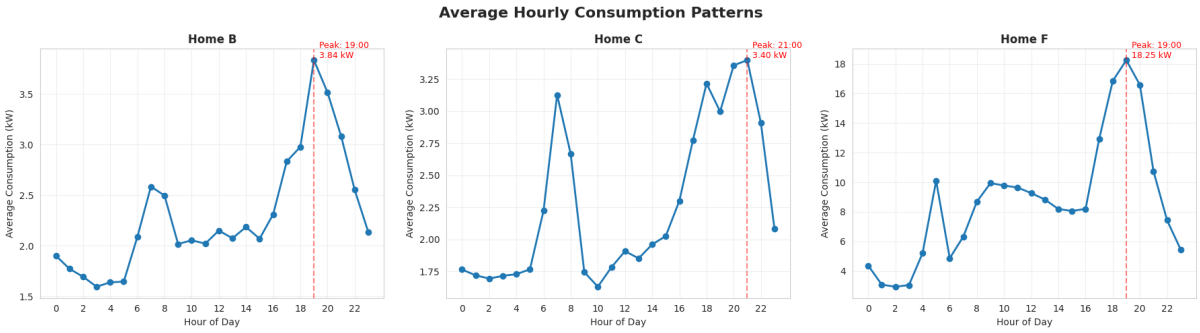


Figure 5.3: Average hourly energy consumption pattern for all homes.

5.2.2 *Data Labeling*

Federated learning with the MLP model follows a supervised learning approach, necessitating labeled data at each client. During local training, clients use these labels to compute prediction errors and update model weights accordingly.However, the datasets utilized in this study are not labeled; therefore, the Isolation Forest algorithm is applied before local training. The model's parameters are configured as follows:

1. Number of estimators (*n_estimators*): Sets the number of trees in the forest and is set to 100 for all homes.

2. Contamination: Specifies the expected proportion of anomalies in the dataset and is distinct for each home.

To determine the appropriate contamination rates, exploratory data analysis is conducted by visualizing the temporal distribution of total energy consumption. Scatter plots of *'Date and Time'*, which describe the time interval, versus *'TotalEnergy'*, which is the cumulative sum of all energy consumption from all devices within a time interval, are generated for each home, as shown in Figures 5.4, 5.5, and 5.6. Based on visual inspection of anomaly patterns, the contamination rate is set to 1.5% for Home B and 1.2% for Homes C and F. The model is then fit to the scaled dataset to identify and predict anomalies. Isolation Forest uses the contamination parameter as a threshold: the specified percentage of points with the lowest anomaly scores are labeled as -1 (anomalies), while the remaining points are labeled as 1 (normal). For clarity and consistency with standard binary classification, these predictions are remapped from $\{1, -1\}$ to $\{0, 1\}$, where 0 represents normal behavior and 1 represents anomalies.This labeling procedure is applied to all homes to ensure consistent anomaly identification across all datasets. Table 5.1 presents the number and percentage of anomalies detected for each home. The detected percentages approximately match the contamination rates specified for each home, indicating that the Isolation Forest model is identifying anomalies in similar to the expected proportions. However, since the dataset is unlabeled, these rates serve as an estimate rather than a definitive measure of true anomalies.

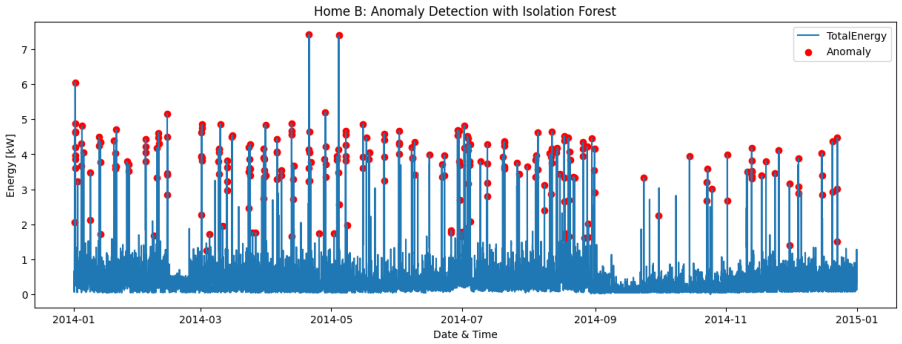


Figure 5.4: Scatter plot of Date vs. Total Energy consumption showing anomalies detected using a 1.5% contamination rate in Home B. Most extreme peaks are identified as outliers, indicating that the chosen contamination percentage is reasonable.

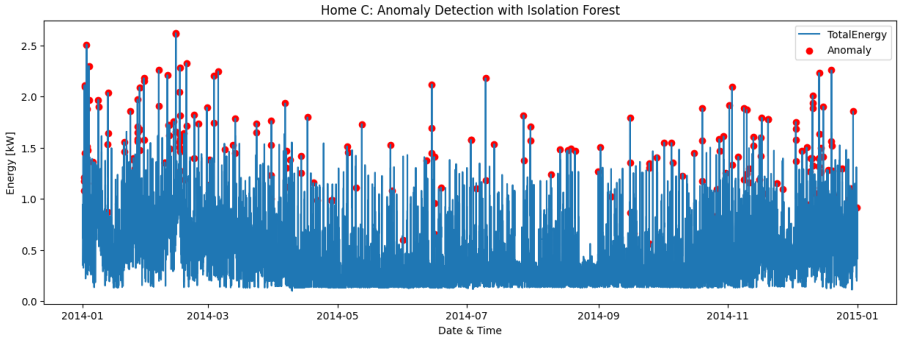


Figure 5.5: Scatter plot of Date vs Total Energy consumption showing anomalies detected using 1.5% contamination rate in Home C.

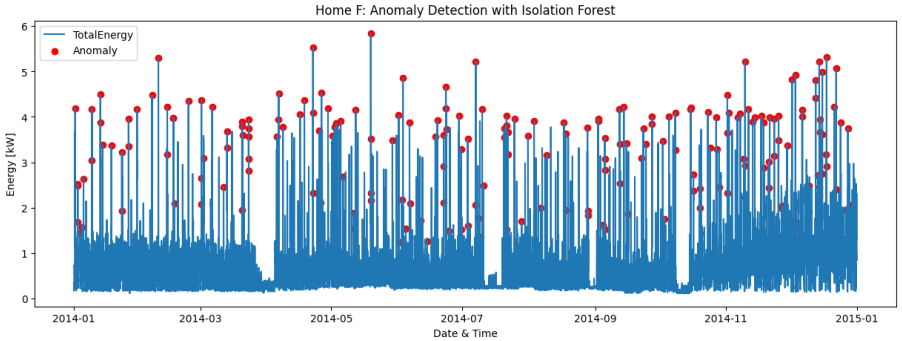


Figure 5.6: Scatter plot of Date vs Total Energy consumption showing anomalies detected using 1.5% contamination rate in Home F.

Home	Number of Anomalies	Percentage of Anomalies
Home B	263	1.50%
Home C	211	1.20%
Home F	211	1.20%

Table 5.1: Anomalies detected by Isolation Forest in each home's dataset

5.2.3 *Local MLP Training:*

Before applying federated learning, the performance of the MLP model trained on each client separately is first tested. This section discusses the architecture of the MLP models and the outcome of anomaly detection on the individual datasets. The MLP architecture starts with an input, taking 7 inputs which are according to the number of devices per home: microwave, fridge, dishwasher, washer-dryer, lighting, and total_energy. The input layer feeds a vector of the features (devices) to the first hidden layer. The first hidden layers is a dense layer consisting of 64 neurons. It receives the input vector, applies ReLU activation function followed by a dropout activation with 0.2 drop probability, setting 20% of neurons randomly to zero which prevents overfitting. The result of the first hidden layer is fed to the second hidden dense layer containing 32 neurons. ReLU activation followed by dropout are also applied in this layer. Finally, the output layer takes the results of the second hidden layer and applies sigmoid activation function which outputs a probability between 0 and 1 describing how likely a sample is anomalous.
Results: Each home is evaluated using the standalone MLP model in two ways:

1. Trained with the same number of local epochs as used in Federated Learning, which is 3.

2. Trained until convergence using early stopping.

Table 5.3 below describes the performance of the MLP model for 3 epochs on each home, same as used during FL. The model is evaluated using four different metrics: accuracy, precision, recall, F1-score. All homes reach very high accuracies reaching an average of 99.38%. However, precision, recall, and F1 measures have varying results with an average of 77.19%, 75.42%, and 75.64% respectively across all homes. The gap between accuracy and other metrics reflect that the model performed poorly as it did not reach its full potential. Table 5.2 presents the results of MLP standalone training until convergence using early stopping. All homes show obvious improvements where each home reached its optimal during different numbers of epochs. For instance, Home B reached convergence during the 14th epoch, Home C at 26, & Home F at 39. Average accuracy across all homes reached 99.6%, precision reached 86.9%, recall 88.2% and F1-score reached 86.71%. Using convergence provides a performance ceiling for each home, representing the best achievable results with local data. By contrast, the 3-epoch training provides a fair comparison to the federated learning setup, where each client performs a limited number of local updates. Reporting both evaluations highlights the benefit of collaborative learning while establishing the standalone baseline under optimal and comparable training conditions.

Metric	Home B	Home C	Home F
Accuracy	99.51%	99.73%	99.62%
Precision	77.08%	100.00%	83.72%
Recall	94.87%	77.42%	92.31%
F1-score	85.06%	87.27%	87.80%
Optimal Epoch	14	26	39

Table 5.2: Standalone MLP Performance Until Convergence for Each Home (Optimal Epochs)

Metric	Home B	Home C	Home F
Accuracy	99.34%	99.57%	99.23%
Precision	72.73%	88.57%	70.27%
Recall	90.57%	73.81%	61.90%
F1-score	80.67%	80.52%	65.82%

Table 5.3: Standalone MLP Performance with 3 Epochs for Each Home

5.2.4 *Federated Learning*

As explained in Chapter 4.3, in this study, FL is used to maintain consumers' private data. Flower (Flwr) , a FL framework, is chosen due to its flexibility, scalability, and ease of use. Federated Learning is carried out for 12 communication rounds. During each round, each client trains its local MLP model for three epochs on mini-batches of 32 samples each. After training MLP models locally, each client sends the learned parameters to the global model. The global model aggregates these parameters by averaging them, then sends them back to each home. During each round, the model progressively improves where the final prediction occurs during the last round. Unlike standalone MLP training, early stopping cannot be applied directly in federated learning. Each client only trains on its local data, and the global model is updated through aggregation after each round. Using too many local epochs can lead to overfitting on individual clients, which reduces the effectiveness of federated averaging. Because of this, the number of local epochs is kept small and set at three to balance enough local learning with stable global aggregation. The number of communication rounds, set at twelve, was chosen based on experience. This allows the global model to converge while keeping communication overhead low. The best number of local epochs and communication rounds is usually determined by the validation performance of the global model across the rounds.

Results: After applying federated learning, the results show steady improvement across the rounds for each client, as presented in Tables 5.4, 5.5, and 5.6. It is important to note that federated learning frameworks like Flower do not support automatic stopping based on convergence. As a result, all clients continue training for the full number of rounds. This means that some intermediate rounds may perform better than later rounds in certain metrics. This effect can be seen in the results. Individual metrics such as precision or recall may peak at different rounds. For example, Client 1 (Home B) achieves its highest precision and recall in round 8 (precision: 90.38%, recall: 88.68%). In contrast, the final round shows slightly different behavior with higher accuracy but lower precision (accuracy: 99.51%, precision: 79.03%, recall:

92.45%). This variation occurs because each metric measures a different aspect of performance, and improvements do not always follow a strict pattern across rounds. Since precision and recall can change independently, the F1 score, which is the average of both, serves as a more balanced measure of overall anomaly detection performance. Clients 2 and 3 reach their highest F1 scores in the last round, while client 1 hits its peak F1 score in round 8. However, the final classification of anomalies or theft for each client is performed using the global model obtained after the last communication round. Figure 5.7 further supports these observations, showing a steady decrease in BCE loss from 0.6658 in the first round to 0.4090 in the final round. This indicates that the global model gradually improves and approaches convergence across communication rounds, even though traditional early stopping cannot be applied in the federated setting.

Round	**Accuracy (%)**	**Precision (%)**	**Recall (%)**	**F1 Score (%**
1	99.32	76.01	92.45	80.3
2	99.34	73.44	88.68	80.34
3	99.46	78.33	88.68	83.19
4	99.51	78.12	94.34	85.47
5	99.57	81.67	92.45	86.73
6	99.57	82.76	90.57	86.49
7	99.46	76.56	92.45	83.76
8	99.69	90.38	88.68	89.52
9	99.63	85.71	90.57	88.07
10	99.57	80.65	94.34	86.96
11	99.57	82.76	90.57	86.49
12	99.51	79.03	92.45	85.22

Table 5.4: Client 1 (Home B) Performance Over 12 Federated Learning Rounds

Round	**Accuracy (%)**	**Precision (%)**	**Recall (%)**	**F1 Score (%**
1	99.37	88.46	54.76	67.65
2	99.49	96.15	59.52	73.53
3	99.51	96.30	61.90	75.36
4	99.57	93.55	69.05	79.45
5	99.57	96.55	66.67	78.87
6	99.54	96.43	64.29	77.14
7	99.60	96.67	69.05	80.56
8	99.54	100	61.9	76.47
9	99.54	100	61.90	76.47
10	99.63	96.77	71.43	82.19
11	99.63	96.77	71.43	82.19
12	99.63	96.77	71.43	82.19

Table 5.5: Client 2 (Home C) Performance Over 12 Federated Learning Rounds

Round	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%
1	99.32	76.47	61.90	68.42
2	99.34	82.76	57.14	67.61
3	99.40	83.87	61.90	71.23
4	99.51	82.05	76.19	79.01
5	99.51	85.71	71.43	77.92
6	99.54	88.24	71.43	78.95
7	99.63	87.18	80.95	83.95
8	99.54	88.24	71.43	78.95
9	99.51	85.71	71.43	77.92
10	99.63	87.18	80.95	83.95
11	99.63	87.18	80.95	83.95
12	99.63	87.18	80.95	83.95

Table 5.6: Client 3 (Home F) Performance Over 12 Federated Learning Rounds

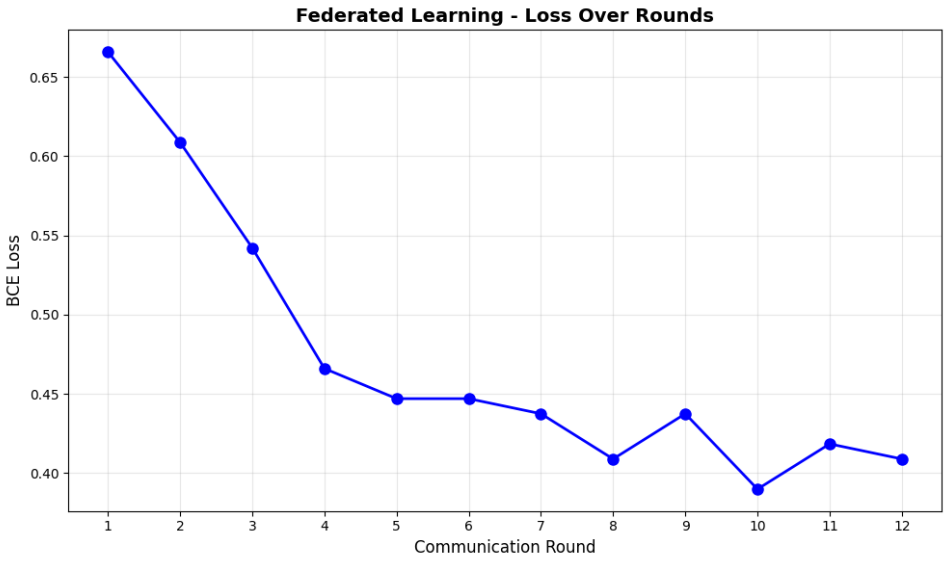


Figure 5.7: BCE loss curve after 12 rounds of federated learning

5.3 Comparison and Analysis

After looking at the performance of anomaly detection, two comparisons were made to evaluate how well federated learning works. The first comparison focuses on standalone MLP models that were trained with the same number of local epochs as in the federated learning setup, which is 3 epochs. The standalone MLP reached an average accuracy of 99% and an average F1-score of 75% across all homes. In contrast, the federated learning approach achieved an average accuracy of 99.6% and an F1-score of 83.7% in the final round. Although both methods use the same local training budget for each round, the federated learning model benefits from

several communication rounds. This allows the global model to progressively improve through repeated aggregation of client updates. On the other hand, the standalone MLP trained for only three epochs does not get to its best performance. Thus, the better performance of the federated learning approach in this case mainly comes from the cumulative learning effect over the rounds, rather than extended local training on individual clients. The second comparison looks at standalone MLP models trained until they converge, using early stopping. In this scenario, the standalone MLP attained an average accuracy of 99.6% and an average F1-score of 86.7% across all homes. Compared to the federated learning results, which had an accuracy of 99.6% and an F1-score of 83.7%, the converged standalone MLP performs slightly better due to the higher number of local training epochs. These epochs are limited in the federated learning setup to avoid overfitting and maintain stable aggregation. This shows that while the proposed Federated Deep Anomaly Detection protects user privacy, it does not always reach the highest level of performance possible with unlimited local training. Overall, these two comparisons offer valuable insights. The first highlights the advantages of federated learning under the same local training budget, while the second shows how it compares to the best standalone models.

Chapter 6

Conclusion and Future Works

In this paper, federated deep anomaly detection (FDAD) framework was proposed for privacy-preserving electricity theft detection in smart meter systems. In this approach, smart meters collect energy consumption data at regular intervals and perform local anomaly detection using an MLP model. Model updates are shared with the utility provider for aggregation, while raw consumption data stays local. This collaborative process repeats over multiple communication rounds, allowing decentralized learning without compromising user privacy. The proposed framework was evaluated against standalone MLP training conducted at each home. Two comparisons were made: standalone training using the same local training epochs as federated learning, and standalone training until convergence with early stopping. While federated learning significantly outperforms standalone MLP models trained with limited epochs, it does not exceed the performance of fully converged standalone models trained independently on each home's data. This result is expected. Standalone training until convergence benefits from unrestricted access to local data and more optimization steps. Federated learning intentionally limits local training to balance communication costs, system stability, and fairness among clients. Importantly, the goal of federated learning is not to outperform fully optimized centralized or standalone models but to enable collaborative learning across distributed data sources in a privacy-preserving and decentralized way.

While a remained performance gap exists between the Federated Learning and the optimal performance of Stand-Alone Models, there are still advantages when using a federated model for large-scale deployments within a Smart Grid Environment, as many cities and regions can use their respective community's knowledge from numerous homes without exposing sensitive residential consumption information. It supports scalable collaboration across thousands of smart meters and reduces privacy risks occurring in centralized data collection. These features make FDAD particularly suitable for real-world smart grid environments, where data privacy, decentralization, and regulatory compliance are as critical as detection accuracy. There is considerable potential to extend this work. More advanced neural network could be explored to further enhance detection performance. Moreover, the proposed method can be further experimented with consumers' datasets that contain different types and numbers of features or appliances, which would provide a more realistic representation of real-world consumption patters in smart meters.

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